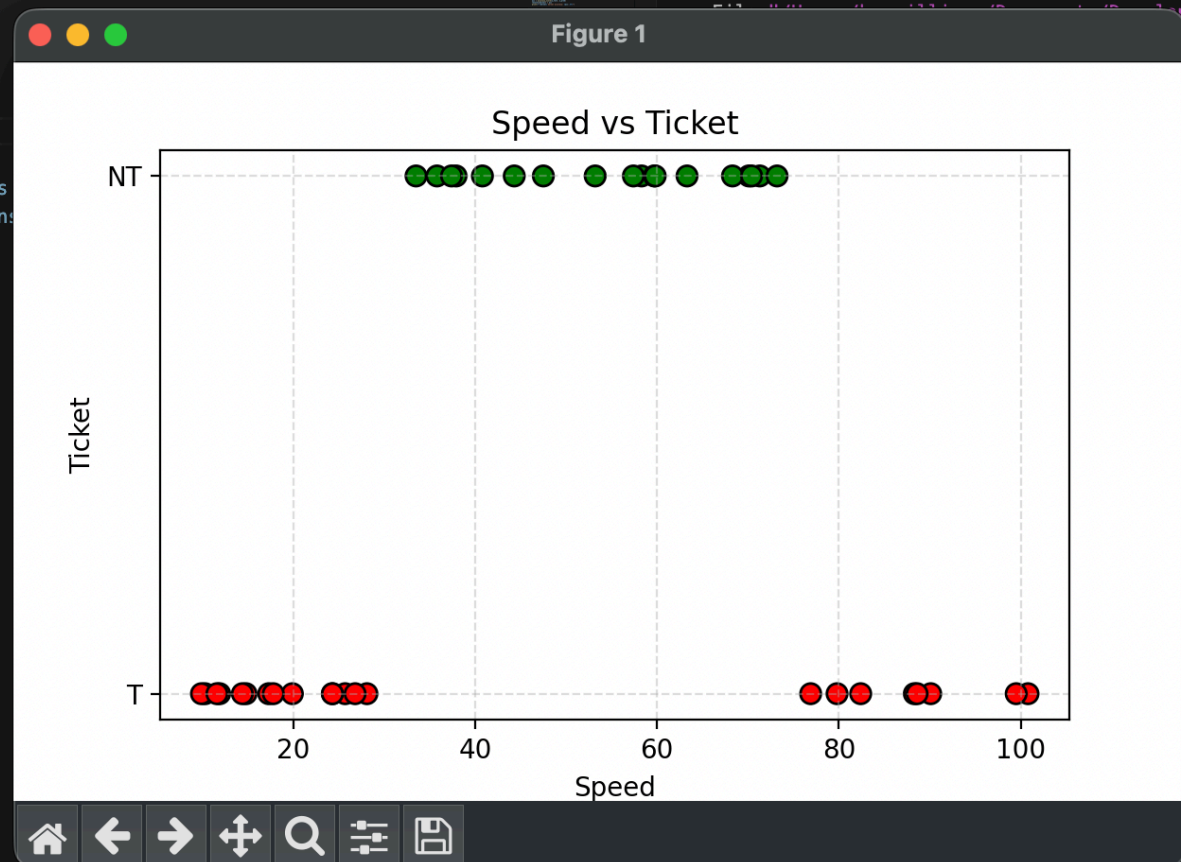


Lab 8 > WilliamsBenjamin_Lab8 > Exercise1 > exercise1.py > ...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 from sklearn.model_selection import train_test_split
4 from sklearn.svm import SVC
5 from sklearn.metrics import accuracy_score
6
7 df = pd.read_csv("speedLimits.csv")
8 df.columns = [col.strip() for col in df.columns]
9
10 speed_col = next((c for c in df.columns if "speed" in c.lower()), df.columns)
11 label_col = next((c for c in df.columns if "ticket" in c.lower()), df.columns)
12
13 df[label_col] = df[label_col].astype(str).str.strip().str.upper()
14
15
16 df = df[df[label_col].isin(["T", "NT"]).copy()]
17
18 y_plot = df[label_col].map({"T": 0, "NT": 1})
19 colors = df[label_col].map({"T": "red", "NT": "green"})
20
21 plt.figure(figsize=(8, 5))
22 plt.scatter(df[speed_col], y_plot, c=colors, s=60, edgecolors="black")
23 plt.yticks([0, 1], ["T", "NT"])
24 plt.xlabel("Speed")
25 plt.ylabel("Ticket")
26 plt.title("Speed vs Ticket")
27 plt.grid(True, linestyle="--", alpha=0.4)
28 plt.show()
29
30 X = df[[speed_col]]
31 y = df[label_col]
32
33 X_train, X_test, y_train, y_test = train_test_split(
34     X, y, test_size=0.1, random_state=0
35 )
36
37 kernels = ["linear", "poly", "rbf", "sigmoid"]
38 results = {}
39
40 for kernel in kernels:
41     model = SVC(kernel=kernel)
42     model.fit(X_train, y_train)
43     y_pred = model.predict(X_test)
44     acc = accuracy_score(y_test, y_pred)
45     results[kernel] = acc
46     print(f"{kernel} kernel accuracy: {acc:.4f}")
47
48 best_kernel = max(results, key=results.get)
49 print(f"\nOptimal kernel: {best_kernel}")
50 print(f"Best accuracy: {results[best_kernel]:.4f}")
```



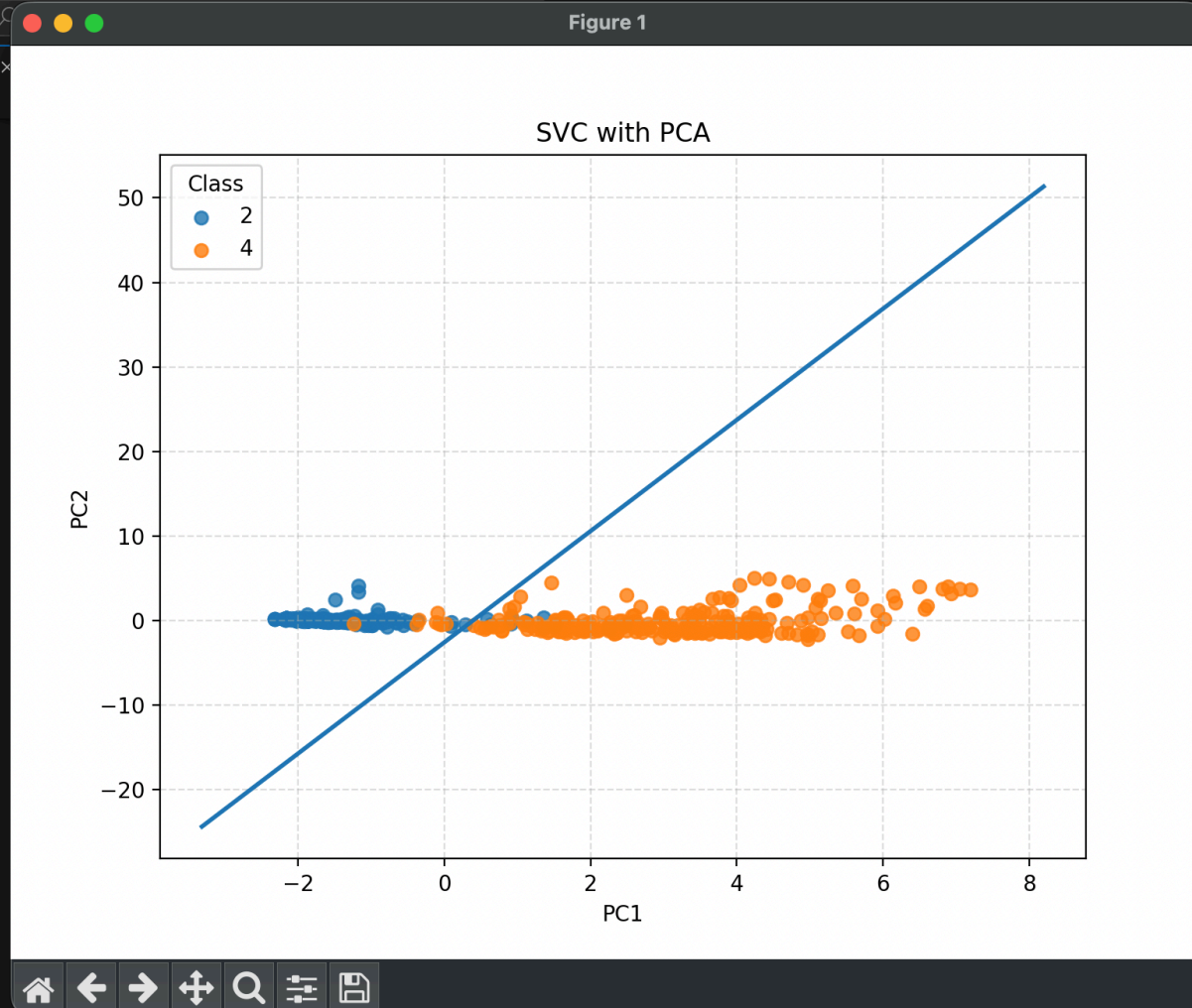
File "/Users/benwilliams/Documents/Development/MachineLearning/.venv/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 300, in _read_parser = TextFileReader(filepath_or_buffer, **kwds)

benwilliams@Mac Lab 8 % cd WilliamsBenjamin_Lab8
benwilliams@Mac WilliamsBenjamin_Lab8 % cd Exercise1
benwilliams@Mac Exercise1 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 8/WilliamsBenjamin_Lab8/Exercise1/exercise1.py"
linear kernel accuracy: 0.5000
poly kernel accuracy: 0.7500
rbf kernel accuracy: 1.0000
sigmoid kernel accuracy: 0.5000

Optimal kernel: rbf
Best accuracy: 1.0000
benwilliams@Mac Exercise1 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 8/WilliamsBenjamin_Lab8/Exercise1/exercise1.py"

```
EXPLORER
...
MACHINELEARNING
> .venv
> .venv-pdf
> .vscode
{} settings.json
> HW
> Lab 1
> Lab 2
  > WilliamsBenjamin_...
  Lab2.pdf
  WilliamsBenjamin_...
> Lab 5
> Lab 7
> Lab 8/William...
  Exercise1
  exercise1.py U
  image.png U
  speedLimits.... U
  Exercise2
  breast-canc... U
  exercise2.py U
> Lab_3
> Lab_4
> Lab_5
> .gitignore
> group4.py U

Lab 8 > WilliamsBenjamin_Lab8 > Exercise2 > exercise2.py > ...
44 X_test_scaled = scaler.transform(X_test)
45
46 pca = PCA(n_components=2)
47 X_train_pca = pca.fit_transform(X_train_scaled)
48 X_test_pca = pca.transform(X_test_scaled)
49
50 model = SVC(kernel="linear")
51 model.fit(X_train_pca, y_train)
52
53 y_pred = model.predict(X_test_pca)
54
55 acc = accuracy_score(y_test, y_pred)
56 cm = confusion_matrix(y_test, y_pred)
57
58 print(f"\nAccuracy: {acc:.4f}")
59 print("\nConfusion Matrix:")
60 print(cm)
61
62 X_scaled_all = scaler.transform(X)
63 X_pca_all = pca.transform(X_scaled_all)
64
65 plt.figure(figsize=(8, 6))
66
67 for cls, color in zip(sorted(y.unique()), ["purple", "gold"]):
68     mask = (y == cls)
69     plt.scatter(
70         X_pca_all[mask, 0],
71         X_pca_all[mask, 1],
72         label=str(cls),
73         alpha=0.8
74     )
75
76 w = model.coef_[0]
77 b = model.intercept_[0]
78
79 x_min, x_max = X_pca_all[:, 0].min() - 1, X_pca_all[:, 0].max() + 1
80 x_values = np.linspace(x_min, x_max, 300)
81
82 if abs(w[1]) > 1e-10:
83     y_values = -(w[0] * x_values + b) / w[1]
84     plt.plot(x_values, y_values, linewidth=2)
85 else:
86     x_line = -b / w[0]
87     plt.axvline(x=x_line, linewidth=2)
88
89 plt.xlabel("PC1")
90 plt.ylabel("PC2")
91 plt.title("SVC with PCA")
92 plt.legend(title="Class")
93 plt.grid(True, linestyle="---", alpha=0.4)
94 plt.show()
```



```
[ 3 65]]
(.venv) benwilliams@Mac Exercise2 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 8/WilliamsBenjamin_Lab8/Exercise2/exercise2.py"
Rows before cleaning: 699
Rows after removing ? : 683
```

Accuracy: 0.9766

Confusion Matrix:

```
[[102  1]
 [ 3 65]]
```